

BHARATH MYSORE

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DISTRIBUTED SYSTEMS AND HIGH PERFORMANCE COMPUTING TOOLS

Systems software engineer and technical leader with 20+ years across cloud infrastructure, distributed storage, HPC and GPU developer tools, and hardware platforms. Combine hands-on C/C++ development with cross-team architecture and delivery from prototype to release. Experience includes Oracle cloud storage, an Azure Storage RDMA prototype, Visual Studio cluster and parallel debugger test teams, and end-to-end HoloLens device integration.

Technical Strengths

Distributed storage and replication; concurrent and parallel programming; HPC/MPI and GPU debugging tools; RDMA; Linux kernel and device drivers; systems root-cause analysis; qualification and deployment automation; engineering mentorship.

Professional Experience

Oracle Corporation | Jun 2016 - Present

Oracle Cloud Infrastructure | Unified Storage and Block Storage

- Design and drive cross-team initiatives for a unified storage platform serving object, block, and file workloads, including shared replication and erasure coding for object storage.
- Build block-storage support on the unified stack and drive migration from the existing backend; contribute to deep archival capabilities. Align work across teams on a new storage platform.
- Lead cross-region and asynchronous volume-group replication from design through delivery for disaster recovery. Ownership spans replication, leader election, snapshots, and backup.
- Scale the backup backend with service growth and implement application-controlled snapshotting. Introduce multiprocessing for iSCSI serving and rearchitect networking to reduce hops and support more attached volumes.
- Increase sellable capacity through cold-data compression, reclaiming 20-30% of storage space. Optimize idle-volume backend capacity through work associated with U.S. Patent 11,412,043; implement trimmed-block reclamation.
- Refactor the backend to contain failures within a feature, volume, or device. Automate failed-disk handling and detection and handling of single-host failures.
- Drive automated qualification and deployments through CI/CD, together with single-click region buildout to reduce manual infrastructure delivery work.
- Design and develop a near-real-time marketplace metering service from prototype to public release; help redirect and implement a data-processing service through internal release.

Professional Experience Continued

Microsoft Corporation | Sep 2005 - May 2016

Visual Studio | Azure Storage | Windows Systems | Device Platforms

- Led Visual Studio Cluster, Parallel, and GPU debugger test teams spanning HPC/MPI, concurrent programming, CUDA, OpenGL, and DirectX. Built local and remote teams, contributed to hiring, and mentored engineers.
- Implemented engine test libraries for cluster and GPU debuggers and architected a reusable debugger UI test library. Worked with vendors to migrate tests across product cycles.
- Designed and implemented an RDMA library for a proof-of-concept Azure Storage backend, applying systems programming expertise to storage transport.
- Owned end-to-end HoloLens depth-camera functionality from prototype through public announcement. Developed hardware-validation drivers and tools for Surface Hub and Kinect.
- Worked on Xbox 360 graphics emulation for Xbox One and a frame-capture tool for collecting graphics metrics under controlled parameters.
- Served as a Windows OS, filesystem, and disk subject-matter expert. Used kernel-dump analysis, reverse engineering, and code review to diagnose disk and cluster failures with customers and development teams.

Intel Technology India Private Limited | Mar 2002 - Sep 2005

Systems Software and Hardware Validation

- Designed and developed CPU-core and cache/memory-subsystem validation tools, including Linux kernel programming and drivers for test cards and chipsets.
- Debugged hardware and operating-system defects, drove issues to closure, and wrote targeted patches. Developed network-card management and configuration software for Windows and Linux.

Earlier Experience

Infineon Technologies (formerly Siemens Semiconductor) | Mar 2000 - Mar 2002. Developed systems software to configure, monitor, and control telephone-exchange cards.

Technical Skills

Languages: C/C++; Java, Python, and C# as needed.

Infrastructure: Kubernetes, Docker, Terraform, Helm, Prometheus, gRPC, Folly, Boost, ZooKeeper, Kafka, and Flink.

Systems: Object, block, and file storage; replication; erasure coding; snapshots; disaster recovery; iSCSI; RDMA; Linux and Windows systems debugging.

Education

B.E., Computer Science and Engineering, Mangalore University